

NITHISH VADUGANATHAN

THIRUVARUR-TAMIL NADU CALL: 8531977455 EMAIL: nithishvaduganathan07@gmail.com
LINKEDIN: [nithishvaduganathan](#) GITHUB: [nithishvaduganathan](#) website: nithishvaduganathan.tech

SUMMARY

Software Developer with strong experience in Python, FastAPI, Flask, REST APIs, and AWS, combined with expertise in AI/ML and Generative AI. Skilled in designing, developing, and deploying scalable backend applications, integrating machine learning models, and building intelligent software solutions. Proficient in Python, SQL, Git, FastAPI, Flask, Scikit-learn, TensorFlow, and cloud technologies. Passionate about writing clean, efficient code and developing high-performance, production-ready applications.

EXPERIENCE

- Operations Intern – Sparktales Learning Solutions Pvt Ltd

11/25 – 03/26

Worked on web-based development and operational tasks, Hands-on exposure to AWS, Azure AI, GenAI, FastAPI, RAG, supported documentation, reporting, and project coordination, Collaborated with team members in a professional environment
- Python & Machine Learning Trainee at JM Consultancy

06/25 – 07/25

Acquired hands-on experience in advanced Python programming, including OOP, file handling, and error handling. Applied machine learning concepts to build mini projects, strengthening foundational ML skills.

PROJECTS

- CITIZEN AI – INTELLIGENT CIVIC ENGAGEMENT PLATFORM:

Built a Flutter mobile app with a Python FastAPI backend for real-time civic issue reporting. Integrated Amazon Cognito, Amazon RDS, AWS EC2, and an AI-powered RAG chatbot for citizen support and civic guidance.
- PRIVACY-PRESERVING AI CHATBOT USING PROMPT MASKING:

Developed a secure AI chatbot using Python FastAPI with prompt masking to detect, mask, and restore sensitive user data before interacting with LLMs, ensuring privacy and compliance.
- SELF-HEALING DRIFT-AWARE MACHINE LEARNING SYSTEM:

Built an ML platform using Python, FastAPI, and Scikit-learn to detect data drift, monitor model performance, and automate retraining and deployment for continuous model accuracy.
- GATE ENTRY MANAGEMENT SYSTEM

Developed a secure gate access management system using **Python, Flask, and SQLite** to automate visitor registration, QR code generation, and entry/exit tracking. Improved security and reduced manual check-in time through real-time visitor verification.

TECHNICAL SKILLS

- Programming languages:

python, java, SQL, dart, C++
- Web development:

React, html, CSS, WordPress, flask, fastApi, restApi, Django, Flutter
- Data science & machine learning:

Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, NLTK, OpenCV, LLMs
- Data Visualization & Analytics:

Tableau, Power BI, Matplotlib, Seaborn, Excel
- Ai Automation tools:

n8n, Zapier, UiPath
- Developer Tools & Version Control:

Git, Jupyter Notebook, PowerShell
- Cloud services:

AWS, Azure, GCP

EDUCATION

- BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

2022 - 2026

SIR ISSAC NEWTON COLLEGE OF ENGINEERING AND TECHNOLOGY NAGAPATTINAM, TAMIL NADU
- HIGHER SECONDARY EDUCATION

2020 - 2022

GOVERNMENT HIGHER SECONDARY SCHOOL, THIRUVARUR, TAMIL NADU

CERTIFICATES

- BCG – GenAI Job Simulation (Forage) — Issued Jun 2025, Credential ID: mx4NjtLsxoyTwysaE
- Tata Group – GenAI Powered Data Analytics Job Simulation (Forage) — Issued Jun 2025, Credential ID: toX97ZvWmxXP8Zbic
- The Fledgling (Infosys) — Issued Mar 2025, Credential ID: 836IW54LX3

ACHIEVEMENTS

- Rotary Youth Leadership Awards (RYLA) – Recognized for leadership development, communication skills, teamwork, and professional networking (Dec 2023).

HOBBIES

- Daily one program
- Open-source contributions
- Blogging on tech/AI trends